

Celestial Mechanics

Problem Set — Solutions

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Section A — Objective

O1. Answer: (b). Directly from Kepler's Second Law, $dA/dt = L/2m = \ell/2$ for any orbit shape.

O2. Answer: (b). The $L^2/2mr^2$ term dominates over $-GMm/r$ as $r \rightarrow 0$, so $V_{\text{eff}} \rightarrow +\infty$ — this is the angular-momentum (centrifugal) barrier, and it's exactly why a planet with nonzero L can never actually reach $r = 0$.

O3. Answer: (c). In AU/yr/ $M_\odot$ units, $T^2 = a^3 \Rightarrow T^2 = 4^3 = 64 \Rightarrow T = 8$ yr.

O4. Answer: (b) is false. L_4 and L_5 are only stable when $M_1/M_2 \gtrsim 24.96$ (option (d)); they are not stable for *every* mass ratio. (a) and (c) are both correct as stated.

O5. Answer: (b). With $\rho_s = \rho_M$, $d_R = R(2)^{1/3} \approx 1.26 R$.

Section B — Subjective

S1.

$$(i) \quad a = \frac{r_p + r_a}{2} = \frac{0.5 + 30}{2} = 15.25 \text{ AU}, \quad e = \frac{r_a - r_p}{r_a + r_p} = \frac{29.5}{30.5} \approx 0.967.$$

(ii) Using $v^2 = GM_\odot(2/r - 1/a)$ with $GM_\odot = 4\pi^2 \text{ AU}^3/\text{yr}^2$ (so 1 AU/yr $\approx$ 4.74 km/s):

$$v_p^2 = 4\pi^2 \left(\frac{2}{0.5} - \frac{1}{15.25} \right) \approx 155.3 \Rightarrow v_p \approx 12.5 \text{ AU/yr} \approx 59 \text{ km/s}.$$

$$v_a^2 = 4\pi^2 \left(\frac{2}{30} - \frac{1}{15.25} \right) \approx 0.043 \Rightarrow v_a \approx 0.21 \text{ AU/yr} \approx 1.0 \text{ km/s}.$$

(Sanity check: $v_p/v_a = r_a/r_p = 60$, consistent with Kepler's Second Law.)

$$(iii) \quad T^2 = a^3 = 15.25^3 \approx 3547 \Rightarrow T \approx 59.6 \text{ yr}.$$

S2.

(i) Radial velocity vanishes at r_p, r_a , so $v = \ell/r$ there, giving

$$\varepsilon = \frac{\ell^2}{2r_p^2} - \frac{GM}{r_p} = \frac{\ell^2}{2r_a^2} - \frac{GM}{r_a}.$$

- (ii) Subtracting the two equations and dividing through by $(r_a - r_p)$ gives $\ell^2 = \frac{2GM r_p r_a}{r_p + r_a}$. Substituting $r_p = a(1 - e)$, $r_a = a(1 + e)$ (so $r_p + r_a = 2a$, $r_p r_a = a^2(1 - e^2)$) gives $\ell^2 = GMa(1 - e^2)$. Combined with $a = -GM/2\varepsilon$, this rearranges directly to

$$e = \sqrt{1 + \frac{2\varepsilon\ell^2}{G^2M^2}}.$$

- (iii) For a circular orbit, $v^2 = GM/r$ gives $\varepsilon = -GM/2a$ (consistent with the general result) and $\ell = \sqrt{GMa}$. Then $2\varepsilon\ell^2/(G^2M^2) = -1$ exactly, so $e = \sqrt{1 - 1} = 0$, as expected.

S3.

- (i) $d_R = R \left(\frac{2\rho_M}{\rho_s} \right)^{1/3} = 6.03 \times 10^7 \left(\frac{2(687)}{920} \right)^{1/3} \approx 6.03 \times 10^7 \times 1.143 \approx 6.9 \times 10^7 \text{ m} \approx 1.14 R$.
- (ii) Using $T^2 = 4\pi^2 d_R^3 / GM$ with $GM \approx 3.79 \times 10^{16} \text{ m}^3/\text{s}^2$ and $d_R \approx 6.9 \times 10^7 \text{ m}$: $T^2 \approx 3.4 \times 10^8 \text{ s}^2 \implies T \approx 1.85 \times 10^4 \text{ s} \approx 5.1 \text{ hours}$.
- (iii) Since $m_s \ll M$ (an icy moon's mass is many orders of magnitude below Saturn's), the barycentre $R_{cm} = (M\vec{r}_M + m_s\vec{r}_s)/(M + m_s)$ sits essentially at Saturn's own centre, so treating Saturn as fixed and using total mass $\approx M$ is an excellent approximation. It would break down only if m_s became a non-negligible fraction of M (comparable to, say, the Moon-Earth mass ratio or larger), at which point the barycentre would noticeably shift away from Saturn's centre.